Formal Mathematical Formulation of Sudoku Puzzle Optimization Problems

1 Mathematical Formulation of Problem B

1.1 Sudoku Grid and Pattern

Let the Sudoku grid consist of 81 cells indexed by

$$i \in \{1, 2, \dots, 81\}, \qquad (r, c),\ r, c \in \{1, \dots, 9\}.$$

A *full solution grid* is a completed Sudoku:

$$G : \{1, \dots, 81\} \to \{1, \dots, 9\},$$

or equivalently a 9×9 array $G[r, c]$.

A *clue pattern* (mask) is a binary vector

$$m \in \{0, 1\}^{81}.$$

Here $m_i = 1$ indicates that cell i is revealed as a clue, and $m_i = 0$ indicates a blank cell.

1.2 Induced Puzzle

Given a pattern m and solution grid G, the induced puzzle $P(m, G)$ is defined cellwise by

$$P(m, G)[i] = \begin{cases} G[i], & m_i = 1, \\ 0, & m_i = 0. \end{cases}$$

1.3 Solution Set of a Puzzle

Let $S(P)$ denote the set of all full Sudoku grids that satisfy every clue in P and all Sudoku rules (row, column, block constraints).

Thus

$$S(P(m, G)) = \{\, X \mid X \text{ is a valid completion of } P(m, G) \,\}.$$

A puzzle is *uniquely solvable* if and only if

$$|S(P(m, G))| = 1.$$

1.4 Difficulty Functional

Let the difficulty score of a puzzle P be defined as

$$\mathrm{Diff}(P) = \alpha \cdot \mathrm{TechDiff}(P) + \beta \cdot \mathrm{BranchDifficulty}(P) + \gamma \sum_{i=1}^{81} \log(|C_i(P)|),$$

where $C_i(P)$ denotes the candidate set of cell i during a logical solving procedure.

1.5 Uniqueness Certification via ILP

The standard Sudoku ILP uses binary decision variables

$$x_{r,c,d} = \begin{cases} 1, & \text{if cell } (r,c) \text{ contains digit } d, \\ 0, & \text{otherwise.} \end{cases}$$

A valid Sudoku solution satisfies

$$\sum_{d=1}^{9} x_{r,c,d} = 1 \quad \forall r, c,$$

and all row/column/block constraints.

Let x^* denote one ILP solution of the puzzle P. Define the index set of its 81 active assignments:

$$I = \{ (r,c,d) \mid x^*_{r,c,d} = 1 \}, \qquad |I| = 81.$$

To forbid this solution, add the constraint

$$\sum_{(r,c,d)\in I} x_{r,c,d} \ \le \ 80.$$

Solving the ILP again with this constraint yields:

$$\text{No new solution exists} \quad \Longleftrightarrow \quad |S(P(m,G))| = 1.$$

1.6 Problem B: Digit Optimization

Fix the clue pattern m. Let $\mathcal{G}$ be the set of all full valid Sudoku grids.

We seek the solution grid G that maximizes puzzle difficulty under pattern m, subject to uniqueness:

$$\boxed{\begin{array}{l} \max\limits_{G\in\mathcal{G}} \mathrm{Diff}(P(m,G)) \\ \text{s.t. } |S(P(m,G))| = 1. \end{array}}$$

This selects the digit configuration that makes pattern m produce the hardest uniquely solvable puzzle.